Specification Document: Intelligent Talent Finder Solution using LinkedIn APIs

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Prepared for: 3rd year students, Department of Computer Science, KITS Warangal

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1. Introduction

1.1 Purpose

This document outlines the specifications for a project aimed at building a smart solution to identify and recommend the right talent from LinkedIn based on specific skill sets or job descriptions. This project will be executed by third-year Computer Science students from KITS Warangal as part of their internship.

1.2 Scope

The solution will leverage official LinkedIn APIs and modern backend and frontend technologies. The goal is to automate parts of the existing talent search process followed by Pi Square Technologies and integrate advanced features into a scalable, efficient system.

1.3 Audience

This document is intended for:
- Interns (3rd-year students), KITS Warangal
- TA team, Pi Square Technologies

2. Project Overview

2.1 Objective

To develop a solution that uses LinkedIn APIs to identify suitable candidates based on a given job description or skill set, without resorting to web scraping.

2.2 Key Goals

- Use LinkedIn APIs for profile access
- Parse job descriptions and extract skill keywords using NLP (Natural Language Processing)
- Automate the shortlisting process using rule-based filtering and machine learning
- Provide a user-friendly interface
- Ensure scalability and maintainability

3. Technology Stack Suggestions

3.1 Backend Technology Options

- Python (Preferred): Flask or FastAPI for RESTful APIs. Includes strong libraries for AI/ML (e.g., Scikit-learn, TensorFlow, spaCy).
- Java: Spring Boot framework. Good for enterprise-grade apps but limited AI/ML library support compared to Python.

3.2 Frontend Technology Options

- ReactJS: Highly interactive UI and suitable for scalable apps.
- Angular: Structured and TypeScript-based.
- Vue.js: Lightweight and beginner-friendly.

3.3 Database Options

- MongoDB (NoSQL): For flexible schema storage.
- PostgreSQL (SQL): For structured relational storage and advanced queries.

4. System Architecture

The architecture includes a frontend single-page application (SPA) built using ReactJS or Angular, a backend REST API in FastAPI (Python), and a database like MongoDB. AI/ML components will be incorporated for:

- Keyword extraction from JDs (NLP)
- Candidate ranking based on AI scoring algorithms
- Recommendation engine for similar profiles

5. Functional Requirements

1. Job Description Input (text upload or manual input)
2. Skill Extraction using NLP
3. Candidate Search via LinkedIn API
4. AI-based Candidate Ranking
5. Profile Summary Display
6. Export Candidate List
7. Dashboard with filters (e.g. location, experience, skills)

6. Non-Functional Requirements

- Fast response time (<5 to 15 seconds per JD)
- API security compliance
- Scalable architecture for multi-user access
- Maintainable and modular code structure

7. Assumptions & Dependencies

- LinkedIn API access is available
- Pi Square will provide sample JDs and existing workflow walkthroughs
- Students will analyze and map the current process for automation

8. Roles & Responsibilities

Students: Design, build, and document the solution
Pi Square: Provide mentorship
KITS Faculty: Facilitate academic guidance and review

9. Milestones & Timeline

1. Requirement Gathering – 1 Week
2. Design & Tech Finalization – 1 Week
3. Development – 4 Weeks
4. Integration & Testing – 2 Weeks
5. Final Demo & Submission – 1 Week

10. Support from Pi Square Technologies

- Share job descriptions and hiring use cases
- Explain current processes
- Weekly sync calls and reviews
- Provide feedback and direction

13. Appendix

- LinkedIn API Docs: https://learn.microsoft.com/en-us/linkedin/
- FastAPI: https://fastapi.tiangolo.com
- Spring Boot: https://spring.io/projects/spring-boot
- NLP Libraries: spaCy, NLTK, Scikit-learn